

PROFESSIONAL SUMMARY

- **8+ years** of experience working in the **healthcare** sector using advanced **Business** and **Data** analytics for data visualization, business intelligence, requirement gathering, Data Analysis, Data Quality assessment, Data Modelling, and information systems.
- Expertise in **ICD** coding and **SNOMED-CT** classification to accurately categorize medical diagnoses and procedures, ensuring compliance with **HIPAA** regulations and enabling precise **data analysis** for **HEDIS** reporting.
- Utilized healthcare standards like **HIPAA**, **HITECH**, **CPT**, **HL7**, and **FHIR** to facilitate uniform **interoperability** and enhance productivity while advancing healthcare operations and data interchange .
- Performed analysis on **clinical outcomes data**, **Medicare and Medicaid claims** data, and **registry data** obtained sources such as **EHRs**, **patient portals**, and **financial applications**.
- Proficient in effectively utilizing the **Epic** electronic health record (**EHR**) system to manage and analyse patient data, contributing to informed clinical decision-making and improved patient care outcomes.
- Proficient in **ICD-9** and **ICD-10** coding systems, ensuring precise medical coding for accurate **claims** processing. Conducted claims analysis, identifying discrepancies and contributing to reduction in coding errors.
- Experienced with managed healthcare plans for **Medicaid** and **Medicare**, **HL 7 standards**, **PPACA**, **LOINC** and **SNOMED**, **GDPR**, **TEFCA**, **FHIR** and **QHIN**.
- Developed workflow and process improvements for the **EMR** using the **Epic EHR software**.
- Proficient in **EHR** data analysis, utilizing advanced analytical tools(**R**, **Python**, **SAS**, **SQL**, **Power BI**, **Tableau**) to uncover actionable insights and support evidence-based healthcare initiatives.
- Experienced in **Agile** and **Waterfall SDLC** environments, involved in requirement gathering, process modelling, and iterative refinement of business requirement documents (**BRDs**).
- Experienced in working with **JIRA software**, including creating and managing projects, workflows, custom fields, dashboards, and reports.
- Engaged in **Agile Scrum** requirement-gathering within the **SDLC** phase, creating **User Stories** through project management software's like **JIRA** and **Azure Devops**.
- Worked on **Use Case Diagrams** and **flowcharts**, aiding Product Owners in defining Functional Requirement Specifications (**FRS**).
- Extensive experience in gathering the **functional** and **non-functional** user requirements for creating the business requirements document (**BRD**) after participating in the Joint Application design (**JAD**) sessions.
- Collaborated with system analysts to develop System Requirements Specifications (**SRS**) with outlines **functional** and **non-functional** requirements of a software system.
- Worked with the Testing team for User Acceptance Testing (**UAT**), implemented **test cases** to validate system functionality.
- Worked with Data Architecture teams in **Snowflake Schema** and **Star Schema** data modelling.
- Proficient in **source-to-target** data mapping for **data migration** from **SQL** to **cloud**, utilizing **ETL** pipelines.
- Worked with **Informatica** for data extraction, transformation, and loading (**ETL**) processes, including data mapping with source qualifier transformation.
- Experience in **Azure SQL server**, conducted monitoring and restoration of **Azure SQL** databases, and successfully migrated the data from **AWS SQL database** to **Azure SQL Server**.
- Proficient in creating intricate ad-hoc **SQL** queries involving Joins, **DML**, **DDL**, **Pivots**, **Views**, **Constraints**, and **functions** across **Oracle**, **MS SQL Server**, and **MySQL**.
- Conducted analysis on more than one million rows of healthcare data to identify healthcare business trends and develop dashboards and stories using **Power BI**, **Tableau**, and **SSRS**.

TECHNICAL SKILLS

Computer skills	Data cleaning, Data Mining, Data Visualization, Microsoft office
Analysis tools	My SQL, Microsoft SQL server, Tableau Desktop, Microsoft Azure, STATA, WEKA, Power BI Desktop.
Regulations:	HIPAA, HITECH, EDI, ICD-10-CM, ICD-10-PCS, HL7, CPT, FHIR, ICH, FDA, PPACA, SNOMED, HEDIS.
Languages	Python, C#.NET, R, SQL, SAS, XML
Cloud	Azure, AWS, GCP
Tools	Git, Jira, Informatica, Azure Devops
Databases	ORACE, MS SQL, MY SQL, SQL server
Operating systems	Windows, Mac OS, Linux

PROFESSIONAL EXPERIENCE:

State of NC

September 2021-till date

Sr. Business Analyst

Responsibilities:

- Throughout the integration of Electronic Health Records (**EHR**) data, strict compliance to **HIPAA** and **HITECH** regulations was maintained, as well as comprehensive data security measures were put in place to protect patient information.
- Successfully integrated healthcare data using **HL7** and **FHIR** standards, enabling seamless interoperability between disparate **EHR** systems and facilitating secure data exchange.
- Led the migration of medical codes from **ICD-9** to **ICD-10**, ensuring accurate representation of diagnoses and procedures and supporting healthcare providers in accurate billing and reporting.
- Implemented **EHR** interoperability enhancements using **Python** and **HL7/FHIR** standards, allowing healthcare providers to access and share patient data more efficiently across different systems.
- Utilized **data analytics** tools to extract insights from **Medicare** and **Medicaid** data, supporting program evaluation and decision-making.
- Practiced **Agile** methodologies, prioritizing user stories and collaborating with cross-functional teams to deliver incremental improvements to the **EHR** integration process.
- Suggested enhancements to existing **SDLC** processes to streamline data analysis workflows.
- Produced comprehensive **BRD** and **FRD** documentation, clearly outlining project scope, requirements, and deliverables, facilitating efficient communication and project management.
- Facilitated effective communication among **cross-functional** teams and stakeholders, ensuring clarity in project requirements and objectives.
- Collaborated with stakeholders to **gather requirements** and ensure alignment between technical solutions and business objectives.
- Played a key role in **Agile project** delivery, prioritizing **user stories**, and actively collaborating with cross-functional teams to deliver incremental improvements in the **EHR** integration process.
- Collaborated with stakeholders to gather and document business and functional requirements (**BRD** and **FRD**), ensuring alignment of technical solutions with business objectives.
- Incorporated user feedback from **UAT** into iterative development cycles, ensuring that the integrated **EHR** system met user needs and requirements.
- Translated business requirements into **user stories** and coordinated User Acceptance Testing (**UAT**) activities to validate system functionality and ensure end-user satisfaction.
- Established **Data governance** practices, including regular compliance audits, to ensure data quality, integrity, and compliance with regulatory standards.

- Led **Data mapping** and transformation efforts, standardizing **EHR** data formats and codes, and enabling accurate analysis and reporting across different healthcare systems.
- Regularly updated the **data dictionary** to reflect changes in data structures, schema modifications, and new data elements.
- Integrated **healthcare data** from diverse systems, including Electronic Health Records (**EHR**), claims systems, and external data sources, into the **data warehousing** for comprehensive analysis.
- Developed and maintained Extract, Transform, Load (**ETL**) processes for **healthcare data**, ensuring the seamless flow of information into the **data warehousing** from various source systems.
- Automated **ETL** workflows using **Python** scripts and **Informatica**, reducing manual intervention and expediting data processing tasks.
- Optimized **SQL** queries to enhance data retrieval performance, resulting in quicker access to critical healthcare information.
- Conducted complex data analysis using **SQL** queries on integrated **EHR** data, extracting meaningful insights that informed medical decision-making and improved patient care.
- Utilized **Python scripting** and **Power BI** for data manipulation, visualization, and analysis, showcasing proficiency in emerging technologies within the information systems industry.
- Utilized **Python** scripting to manipulate and clean **EHR** data, improving data quality and facilitating more accurate analysis and reporting.
- Designed interactive dashboards in **Power BI** to visualize key healthcare metrics, providing healthcare professionals with real-time insights to enhance patient care and operational efficiency.

Centene corporation, Saint Louis, MO

January 2019 – July 2021

Business Analyst

Responsibilities:

- Demonstrated strong logical reasoning and analytical abilities through orchestrating end-to-end **HIPAA**-compliant **data integration** and harmonization of Electronic Health Records (**EHR**) and Electronic Medical Records (**EMR**) using **FHIR** and **HL7** standards.
- Effectively communicated project objectives, protocols for **HIPAA** data access controls, and **data governance** efforts to stakeholders, ensuring patient data confidentiality.
- Expertly mapped **source-to-target** data flow, ensuring accuracy and consistency of data across **EPIC** systems, adhering to **ICD-9** and **ICD-10** coding standards.
- Constructed **HIPAA**-compliant data models using **snowflake** schema design within an enterprise **data warehousing** framework, optimizing data storage and retrieval.
- Created interactive **Tableau** visualizations to present **ICD-9** and **ICD-10** coding trends, enhancing reporting accuracy and facilitating informed clinical decisions.
- Ensured compliance with **Medicare** and **Medicaid** regulations, including the Centers for **Medicare & Medicaid** Services (CMS) guidelines and state-specific requirements.
- Championed **Agile** principles through **Scrum** practices, collaborating on sprint planning, backlog grooming, and iterative development using **Jira**.
- Led a multidisciplinary team in **agile** environments, utilizing **Scrum** methodologies and **Jira** for streamlined project management of **HL7** data integration, resulting in accelerated **SDLC** cycles.
- Played a pivotal role in **SDLC** improvement initiatives, proposing data-centric enhancements during **Scrum** retrospectives for increased project efficiency.
- Facilitated Joint Application Development (**JAD**) sessions to elicit requirements, translating them into robust Business Requirements Documents (**BRD**) and detailed Functional Requirements Documents (**FRD**).
- Produced comprehensive Business Requirements Documents (**BRD**) and detailed Functional Requirements Documents (**FRD**) to facilitate clear communication and project management.
- Conducted thorough User Acceptance Testing (**UAT**) to validate data integrity and system functionality, ensuring seamless **EPIC** integration and compliance.

- Generated detailed **SAS** reports, conducting statistical analysis on patient data to identify patterns and trends, contributing to proactive healthcare strategies.
- Conducted thorough analysis of **Medicare** and **Medicaid** benefit plans, assessing their impact on beneficiary access, quality of care, and cost containment.
- Created and applied **data validation** rules and checks to identify and rectify discrepancies, improving **data reliability** and **usability**.
- Designed and implemented automated **scripts** and workflows to perform routine **data validation** tasks, reducing manual effort and increasing efficiency.
- Translated logical **data models** into physical data models, optimizing for performance, storage, and **database** technology constraints.
- Designed and implemented dimensional models like **Snowflake schemas** for **data warehousing** and analytical purposes.
- Managed **version control** for **data models** and participated in ongoing model maintenance and updates.
- Collaborated with data engineers to optimize **ETL** pipelines, reducing data processing time and enhancing overall data flow efficiency.
- Collaborated in cross-functional teams to design, develop, and deploy insightful reports using **SSRS**, promoting data-driven decision-making across the organization.
- Leveraged advanced **SQL** skills in **SSMS** to execute complex data joins, create optimized views, and design efficient functions for comprehensive health data analysis.
- Proficiently executed complex **SQL** queries, views, and stored procedures in **SSMS**, facilitating in-depth data analysis and reporting on **EMR/EHR** data.
- Introduced **Hadoop** framework to accommodate large-scale data processing and integrated **Azure** and **AWS** cloud services to enhance healthcare **data scalability** and accessibility, showcasing familiarity with emerging technologies.
- Integrated **Azure** and **AWS** cloud services to enhance healthcare data scalability and accessibility, aligning with the organization's **data warehousing** needs.
- Utilized **Python** libraries including **NumPy** and **Pandas** for exploratory **data analysis** and **integrated** advanced **SQL** skills in **SSMS**, demonstrating familiarity with software technology in **data analysis**.
- Developed dynamic **Power BI** dashboards for visualizing healthcare **KPIs**, empowering stakeholders to monitor **HEDIS** metrics and make data-driven decisions.
- Delivered comprehensive training sessions on **Tableau**, **Power BI**, and **Python** for non-technical stakeholders, enabling enhanced data-driven decision-making.

Oscar Health, NYC

September 2016 – December 2018

Business Analyst

Responsibilities:

- Led **HIPAA** compliant data integration project, ensuring secure **EMR-EHR** data exchange and adherence to standards.
- Implemented **HL7** and **FHIR** standards for seamless **interoperability** between different healthcare systems, resulting in improved data sharing and patient care coordination.
- Effectively communicated project objectives, **data exchange requirements**, and **HEDIS** metrics through compelling **data visualizations**, enhancing patient care strategies and informed decision-making.
- Orchestrated successful transition from **ICD-9** to **ICD-10** coding system, minimizing errors and ensuring accuracy.
- Implemented fraud detection measures within **Medicare** and **Medicaid** systems to identify and prevent fraudulent claims and abuse.
- Employed **Agile** methodologies and **Jira** for efficient project management, ensuring on-time **data integration** solutions.
- Collaborated with healthcare providers to define Business Requirements (**BRD**) and Functional Requirements (**FRD**) for data exchange, utilizing **Agile** methodologies.
- Collaborated with healthcare providers to define Business Requirements (**BRD**) and Functional Requirements (**FRD**) for **data exchange**, showcasing strong communication skills in translating healthcare needs into technical solutions.

- Utilized **Agile** methodologies in developing **UAT** strategies and collaborating with healthcare providers, highlighting experience with **Agile** practices.
- Developed **UAT** strategies, created **data validation** scripts in **SAS**, and leveraged data warehousing techniques.
- Enhanced **data integrity** by applying rigorous **data validation** techniques, leading to reduction in discrepancies.
- Demonstrated how the use of the **data dictionary** led to improved data understanding, consistency in reporting, and enhanced data-driven decision-making.
- Designed **snowflake** schemas and executed **ETL** with **Informatica**, and reduced **data discrepancies**.
- Integrated diverse **data sources** into a unified analytics platform, enabling comprehensive and cohesive analysis.
- Crafted historical data snapshots using **data warehousing** principles for insightful trend analysis.
- Created **data validation** scripts in **SAS** and **SSRS**, systematically identifying discrepancies and ensuring coding precision.
- Conceptualized and designed both **star** and **snowflake** schemas, optimizing **data retrieval** speed and analytical efficiency.
- Developed **SQL** queries, joins, views, and functions in SQL Server Management Studio (**SSMS**) to facilitate **data extraction** and **reporting**.
- Produced impactful reports in **SSRS**, applied **R-Programming** for data analysis and modelling.
- Utilized **Azure** for hosting interactive healthcare analytics dashboards in **Power BI**, aligning with the organization's data architecture needs.
- Developed interactive healthcare analytics dashboards in **Power BI**, integrated data sources, and used **Azure** for hosting.
- Utilized **Python libraries** like **NumPy**, **Pandas** for **data pre-processing**, collaborated with coders for accurate **ICD-9** to **ICD-10** migration.

Bluebird Bio Health, Durham, NC

May 2015 – August 2016

Business Analyst

Responsibilities:

- Successfully integrated healthcare data from various sources, ensuring compliance with **HIPAA** regulations and utilizing **HL7**, **FHIR**, and **Epic EHR** systems.
- Significantly improved the reporting quality for **HEDIS** indicators, which led to improved **Medicare** and **Medicaid** reimbursement processes.
- Spearheaded **source-to-target data mapping** efforts, achieving precise and efficient data transfers while streamlining data cleaning processes.
- Employed **Waterfall SDLC** methodology to oversee project stages, from requirement analysis to User Acceptance Testing (**UAT**), resulting in on-time and on-budget project delivery.
- Collaborated with stakeholders to gather requirements and facilitated Joint Application Development (**JAD**) sessions to create comprehensive Business Requirements Document (**BRD**) and Functional Requirements Document (**FRD**).
- Assisted in the design and execution of **UAT**, ensuring data accuracy and system functionality alignment with user needs.
- Utilized Informatica **ETL** tool to design and implement complex **ETL** workflows for **data extraction**, transformation, and loading, ensuring consistent **data validation** and **data integration**.
- Developed and maintained **Snowflake schema**-based **data models**, optimizing **data warehouse** performance and enabling efficient data retrieval for analysis.
- Cleaned, validated, and transformed raw data using **SAS**, ensuring data accuracy and consistency for further analysis.
- Utilized **SQL** and SQL Server Management Studio (**SSMS**) to create advanced SQL queries, joins, views, and functions for data analysis and reporting purposes.
- Collaborated with cross-functional teams to leverage **Azure** cloud services for data storage, integration, and analysis, utilizing **Power BI** for intuitive **data visualization**.
- Produced insightful reports using **MS Excel**, **spreadsheets**, and **Power BI** dashboards, providing actionable insights to stakeholders.